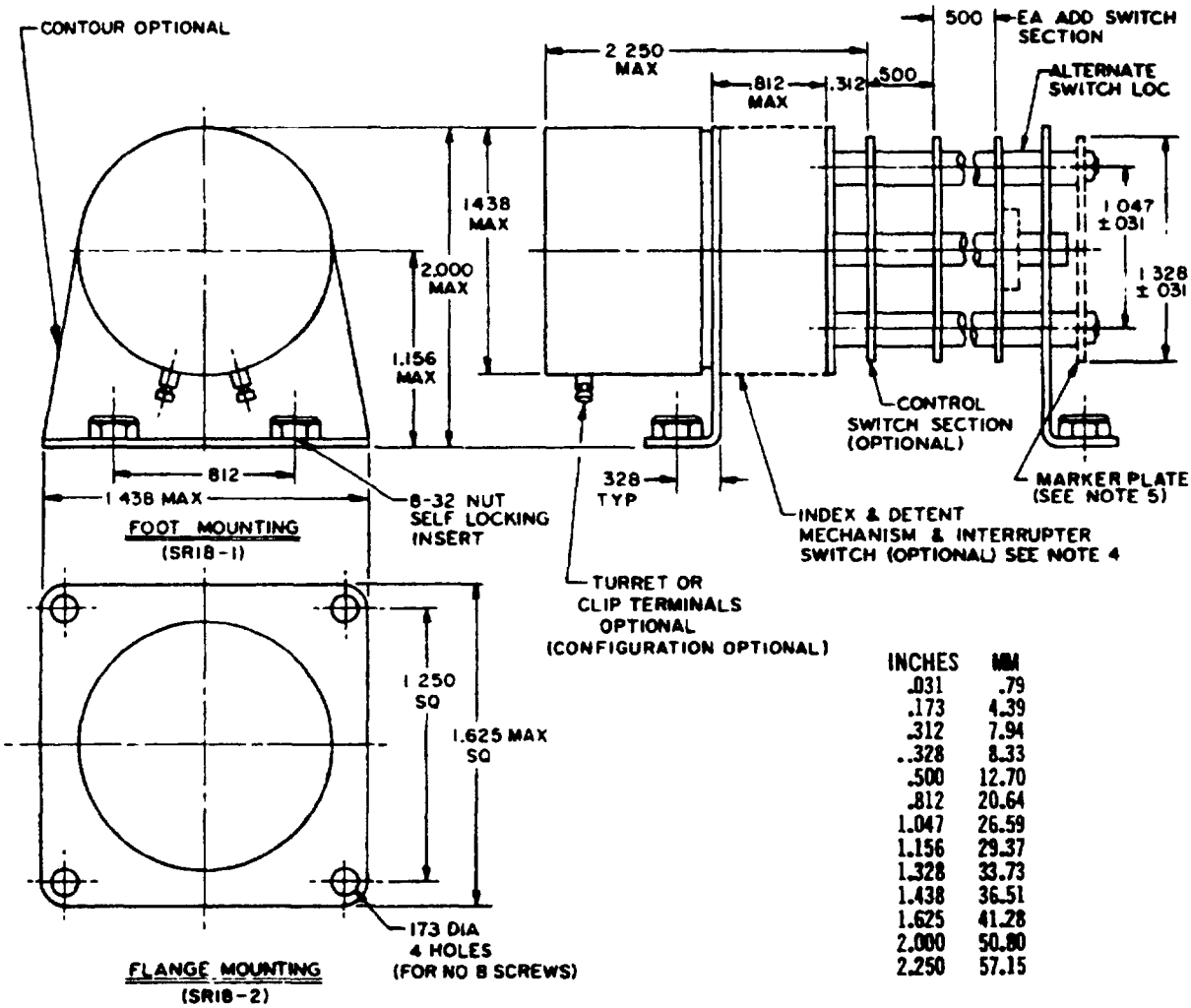
MILITARY SPECIFICATION SHEET

SWITCH, ROTARY, ELECTRO-MECHANICAL ACTUATED (SOLENOID TYPE), OPEN CONSTRUCTION, 1/2 AMPERE, STYLE SR18

The complete requirements for procuring the switch described herein shall consist of this document and the latest issue of Specification MIL-S-3786.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



INCHES	MM
.031	.79
.173	4.39
.312	7.94
.328	8.33
.500	12.70
.812	20.64
1.047	26.59
1.156	29.37
1.328	33.73
1.438	36.51
1.625	41.28
2.000	50.80
2.250	57.15

NOTES
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.015 (.38 mm)
4. Interrupter switch mechanism may extend below dotted line, but not below mounting feet.
5. Marker plate may be omitted if rear foot bracket is visible and contains markings.

FSC 5930

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REQUIREMENTS:

Dimensions and configuration· See figure.

Metals: Metals shall be of a corrosion-resistance type, or shall be cadmium plated in accordance with class 3, type II, of QQ-P-416.

Terminals Terminal tabs shall be bent at an angle of 75 ± 15 degrees or 0 ± 15 degrees, as specified.

Solenoid-input voltage: Solenoid-input voltage shall be 24 vdc minimum, 30 vdc maximum.

Solenoid-input current· Maximum-input current shall not exceed 3-1/2 amperes at 85° C or 3 amperes at 125° C at 26 vdc.

Life (rotational). Test loads and applicable circuit conditions shall be as specified in table I. Each of the loads specified for the applicable environmental condition shall be switched by at least one rotor contact of the switch.

TABLE I. Electrical loads.

Environmental condition	Inductive load (2.8 Henries)		Resistive loads	
	Milliamperes	Volts, dc	Milliamperes	Volts
At atmospheric pressure	50	28	500 225 50	28 dc 115 vrms 300 vrms
At reduced barometric pressure	---	---	350 100	28 dc 115 vrms

Dielectric withstanding voltage. The applicable test voltage specified in table II shall be applied between the general switch elements.

Stepping speed: Solenoid-coil current shall be interrupted at a rate sufficient to cause switch deck contacts to be stepped at the rate of 20 steps per second, minimum.

TABLE II. Dielectric-test voltages.

Altitude	Test voltage (Volts, rms)
At atmospheric pressure	1,000
At reduced barometric pressure	400

Contact resistance· For switches of life-temperature characteristics A, C, and E, the contact resistance at room temperature shall not exceed the following values.

Initial and after vibration and shock - - - - 20 milliohms.
After moisture resistance and salt spray - - - - 30 milliohms.
After initial soak and completion of life test - - - - 40 milliohms.

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Ordering data: Procurement documents shall specify the following

(a) Angle of bend for terminal tabs (see 3.2.1).
(b) Spacer dimensions if greater than 1/2 inch.
(c) Temperature characteristic
(d) Life characteristic.
(e) Flange or foot mounting (see figure).
(f) Self-interrupter switch, as required.
(g) Homing (control) deck and configuration, as required.
(h) Number of switch positions, angle of throw, and electrical configuration.
(i) Solenoid input voltage range and power source type.
(j) Solenoid duty cycle. Specify maximum "ON" and minimum "OFF" times.
(k) Lead wires and length (when required).

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU, MI
Navy - EC, AS, OS
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME, AT, AV, SM
Navy - MC, CG
Air Force - 19

Preparing activity:
Navy - EC

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